AGV2 Kayaba — Function Summary

Differential-drive magnetic-tape AGV with front and rear MGS1600 sensors, RFID route tags, trolley-hitch pusher, lidar safety fields and a Flask web HMI. Read from `profiles/agv2_kim.json` and the controller code at commit 7496168 · 5 Oct 2026. All RFID tag values are placeholders until replaced with the real scanned hex.

1. Operating modes

Mode	Entered by	Behaviour
ARMED	Startup, RESET, arriving at homepost or B0	Stopped, motors released, waiting for a start command.
RUNNING (AUTO forward)	START (DI0), CALL START (DI4), branch button (DI5–8), or end of a reverse	Forward PID tape-following on the front MGS1600.
REVERSING	Tag 0001, or web START REVERSE	Backward PID tape-following on the rear MGS1600.
MANUAL	Mode selector DI2 ON	Jogging from the web remote (pendant inputs not wired).
EMERGENCY	E-stop DI3	All motion stops, motors released (AGV can be pushed). RESET to recover.

2. Ways to start

- **START (DI0)** — normal AUTO run from ARMED. Needs tape under the front sensor.
- **CALL START (DI4)** — only at homepost (DO11 ON). That run stops at tag B0; press START to continue.
- **Branch buttons (DI5–8)** — in ARMED: selects TITIK 1–4 **and** starts AUTO (normal run). Ignored while driving.
- **Web START REVERSE** (MANUAL page) — starts REVERSE from ARMED, needs tape under the rear sensor. Web STOP returns to ARMED.

3. Speeds

Mode / setting	Value	Notes
SLOW (tag 0014)	0.2 m/s	SLOW PID gains
MEDIUM (tag 0015)	0.4 m/s	SLOW PID gains
HIGH (tag 0016)	0.6 m/s	HIGH PID gains
MAX (tag 0018)	1.0 m/s	HIGH PID gains · motors top out at about 0.94–0.98 m/s
Reverse	0.15 m/s	0.1 m/s after tag 00CA · 30 s safety timeout (paused while holding)
AUTO accel / decel	0.2 / 0.2 m/s²	ACCEL_RATE / DECEL_RATE
Manual high / slow	0.4 / 0.2 m/s	Slow speed used for turning
Manual accel / decel	0.8 / 0.8 m/s²	MANUAL_ACCEL_RATE / MANUAL_DECEL_RATE
Lidar outer / middle cap	0.5 / 0.2 m/s	Eases down at 0.2 m/s²

- Forward steering direction is flipped (`ERROR_SIGN_FORWARD: 1`).
- Tape loss or sensor (CAN) timeout: brake, resume automatically when the tape is seen again.
- Speeds, ramps and PID gains can be tuned live on the PARAMS page (RAM only; restart restores the profile).

4. RFID mapping

Tag	Sequence	Fires when	Action	Cooldown
0014	speed_zone_slow	RUNNING	Speed → SLOW (0.2 m/s)	0 s
0015	speed_zone_medium	RUNNING	Speed → MEDIUM (0.4 m/s)	0 s
0016	speed_zone_high	RUNNING	Speed → HIGH (0.6 m/s)	0 s
0018	speed_zone_max	RUNNING	Speed → MAX (1.0 m/s)	0 s
000C	checkpoint_stop	RUNNING	Stop 4 s, resume at SLOW	0 s
00AA	branch_titik_1	RUNNING + branch 1	SLOW, track left	5 s
00AB	branch_titik_2	RUNNING + branch 2	SLOW, track right	5 s
00AC	branch_titik_3	RUNNING + branch 3	SLOW, track left	5 s
00AD	branch_titik_4	RUNNING + branch 4	SLOW only (no track action yet)	5 s
00AE	fork_always_right	RUNNING (every run)	Track left	5 s
00BB	branch_merge_reset	RUNNING (every run)	Track right, clear selected branch	5 s
0001	reverse_at_1	RUNNING, skipped if ignore_rfid_1 set	Stop, wait 2 s, start REVERSE	3 s
00CA	reverse_approach_ca	REVERSING	Reverse speed → 0.1 m/s	3 s
00CC	stop_at_c1	REVERSING (also DI10 marker)	Hitch sequence — see section 6	5 s
00CE	clear_ignore_at_cr	Any mode	Clear ignore_rfid_1 (tag 0001 works again)	3 s
0000	arrive_homepost	RUNNING, not right after leaving home	Stop, exit to ARMED, DO11 ON	5 s
00B0	stop_at_b0	RUNNING, CALL START runs only	Stop, exit to ARMED	5 s

5. IO mapping

DIO module 192.168.1.31 (Modbus TCP, 16 DI / 16 DO). All DIs are inverted in software (`DI_FLIPPED = 1`): “ON” below is the value the code sees, and an unwired input reads ON.

Digital inputs

DI	Function	Meaning / use
0	START	Rising edge = start AUTO from ARMED
1	RESET	Rising edge = reset / leave emergency
2	MODE selector	ON = MANUAL, OFF = AUTO
3	EMERGENCY	ON = e-stop tripped
4	CALL START	Start AUTO only at homepost; run stops at B0
5	BRANCH 1 button	Select TITIK 1 (+ start in ARMED)
6	BRANCH 2 button	Select TITIK 2 (+ start in ARMED)
7	BRANCH 3 button	Select TITIK 3 (+ start in ARMED)
8	BRANCH 4 button	Select TITIK 4 (+ start in ARMED)
9	TROLLEY LIMIT 1+2 (NC)	Both switches in parallel · currently ignored
10	REVERSE STOP MARKER	Rising edge while REVERSING = precise stop
11	LIDAR OUTER	ON = object · cap 0.5 m/s (AUTO forward)
12	LIDAR MIDDLE	ON = object · cap 0.2 m/s (AUTO forward)
13	LIDAR INNER	ON = object · stop at once (AUTO forward)
14	PUSHER PROX TOP	Pin at top = “hooked”

DI	Function	Meaning / use
15	PUSHER PROX BOTTOM	Pin at bottom · ends FULL DOWN

Digital outputs

DO	Function	Meaning / use
0	LEFT driver FWD	Direction forward
1	LEFT driver REV	Direction reverse
2	LEFT driver BRAKE	Brake
3	LIDAR AREA SEL 1	Lidar field set (set_lidar_area), OFF by default
4	RIGHT driver FWD	Direction forward
5	RIGHT driver REV	Direction reverse
6	RIGHT driver BRAKE	Brake
7	LIDAR AREA SEL 2	Lidar field set (set_lidar_area), OFF by default
8	HORN AUTO	ON while RUNNING or REVERSING
9	HORN ALARM	Follows alarm_active (nothing triggers it yet)
10	spare	—
11	AT HOMEPOST	ON while parked at homepost
12	PUSHER DOWN a	Relay K1 · with DO15
13	PUSHER UP a	Relay K2 · with DO14
14	PUSHER UP b	Relay K3 · with DO13
15	PUSHER DOWN b	Relay K4 · with DO12

Analog outputs (AO module 192.168.1.30, register base 4096)

AO	Function	Range
0	LEFT motor speed	Code sends 0–5 V (driver 0–10 V → 0–3000 rpm)
1	RIGHT motor speed	Code sends 0–5 V (driver 0–10 V → 0–3000 rpm)

Network and CAN

Device	Link	Address
IPC (Ubuntu)	Ethernet	192.168.2.20 · web HMI on port 5000
DIO module	Modbus TCP	192.168.1.31:502 · unit 1
AO module	Modbus TCP	192.168.1.30:502 · unit 1
RFID reader	TCP	192.168.3.200:2022
MGS1600 front	CAN 500 kbit/s (CANable2 USB)	Node 6 · TPDO 0x186
MGS1600 rear	CAN 500 kbit/s (CANable2 USB)	Node 7 · TPDO 0x187 · REVERSE only

6. Reverse and trolley-hitch sequence

Step	Trigger	What happens
1	Tag 0001 (RUNNING)	Stop, wait 2 s, switch to REVERSE (skipped while ignore_rfid_1 is set).
2	—	Reverse at 0.15 m/s on the rear sensor (always the right track).
3	Tag 00CA	Slow to 0.1 m/s for the approach.
4	DI10 marker or tag 00CC (first wins)	Brake and hold, still in REVERSE. Set ignore_rfid_1.
5	+2 s	Pusher UP automatically (DO13 + DO14).
6	DI14 ON (max 30 s)	Leave REVERSE, drive forward at SLOW. On timeout it continues forward anyway.
7	—	Tag 00CC ignored for 5 s after moving off; tag 0001 ignored until tag 00CE.

7. Pusher (trolley hitch)

- UP = DO13 + DO14, DOWN = DO12 + DO15, through a 4-channel relay board wired as an H-bridge. The UP and DOWN pairs are never energised together by the code.
- Web FULL UP / FULL DOWN stop at DI14 / DI15 or after 15 s. JOG UP / DOWN ignore both proximity sensors. PUSHER E-STOP latches until cleared.
- At startup all pusher outputs are forced OFF; the pin is not moved. Lowering is always manual.
- Trolley limit switches (DI9) are ignored for now (`IGNORE_TROLLEY_LIMIT: true`): “hooked” = pin at DI14.

8. Lidar safety fields (AUTO forward only)

Input	Object in field	Effect
DI11 outer	ON	Ease down to at most 0.5 m/s
DI12 middle	ON	Ease down to at most 0.2 m/s
DI13 inner	ON	Brake immediately; when clear, wait 2 s, then continue

- The lidar only lowers speed; it never raises it above the tag speed.
- DO3 / DO7 select the lidar field set via the `set_lidar_area` action on RFID tags (none assigned yet; both OFF by default and after reset).
- Not active while REVERSING or in MANUAL.

9. Manual mode

- Web remote jog: forward/reverse at 0.4 m/s, turns and spins at 0.2 m/s.
- Smooth start and stop at 0.8 m/s²; a direction change slows through zero first. E-stop still cuts power immediately.

10. Homepost, horn and status outputs

- DO11 AT HOMEPOST: ON while parked at homepost. At power-up the AGV is assumed to be at homepost (`ASSUME_AT_HOME_ON_BOOT`).
- DO8 horn AUTO: ON while RUNNING or REVERSING. DO9 horn ALARM: reserved.

11. Web dashboard (http://<ipc>:5000)

Page	Functions
HOME	Mode, sensors, PID values, RFID, branch selector (TITIK 1–4), sequence status
MANUAL	Jog pad, pusher panel (FULL UP/DOWN, JOG, E-STOP), AUTO REVERSE start/stop
IO	Live DI/DO lamps; IO TEST mode to force each DO ON/OFF (auto-exits when the AGV drives)

Page	Functions
PARAMS	Live tuning of speeds, ramps and PID gains (RAM only)
ERRORS	Error log, controller restart button

12. Open items

Note
• Replace all placeholder RFID tag values with the real scanned hex.
• TITIK 4 (00AD) has no track action yet; after 00CC the AGV keeps the last selected track.
• Set lidar `ENABLED: false` until the lidar is wired (unwired inputs read ON → inner field → AGV will not move in AUTO).
• Pusher relay protection not added yet: UP/DOWN interlock on the IO test page and a dead-time on direction change. Fit a fuse on the relay supply.
• MAX 1.0 m/s is above the motor's ~0.94 m/s capability and leaves no steering headroom.
• Spare channels: DO10 only. agv1_kim.json does not load (missing DECEL_RATE).